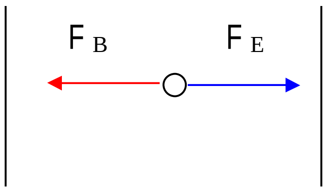
4.(a) (i)



(ii) The alpha particle would be deflected to the **right** along a **parabolic path.**

As the speed of the particle is less than v, the **magnetic force experienced is less than the electric force.** Hence the particles experience **a constant acceleration to the right.**

(b) Stage 2

By Newton's 2nd law, the magnetic force provides the centripetal force for electron to undergo circular motion

$$Bqv = mv^2/r$$

$$\Rightarrow r = mv/Bq \text{ ------ } (1)$$

$$= (4 \times 1.67 \times 10^{-27} \times 1.00 \times 10^5)/(0.050 \times 2 \times 1.6 \times 10^{-19})$$

$$= 4.175 \times 10^{-2} \text{ m}$$

Point of impact = $2r = 2 \times 4.175 \times 10^{-2}$ m $= 8.35 \times 10^{-2}$ m

(c) Stage 1:

As the Beta particle velocity is 100 times faster, The magnetic force is now 100 times stronger than the electric force. Hence, an electric field strength 100 times the original one must be applied for it to remain undeflected.

For alpha particle,

$F_E = F_B$

$qE = Bqv$

$v = E/B$

Since beta particles travels 100 times faster,

$F_B' = B(2q)(100v)$

$F_E' = (2q)\ E'$

For Beta particle to remainl undeflected,

$F_B' = F_E'$

E' = 100Bv = 100 E

Stage 2:

- While the speed of beta particles is 100 times faster, its mass is more than 1000 times smaller than alpha particle. Hence the position of photographic plate needs to be placed much closer to the entrance since the point of impact is much smaller now.
- Beta particle is negatively charge, the point of impact is on the right side of the entrance. Thus the position of the plate must be placed at the right side of the entrance instead of the left side.

From (1),

Radius of He particle, r = *mv/Bq*

Radius for Beta particles, r' = $m_e v_e / B2q$

$$= \frac{\frac{1}{1833} m\,(100v)}{2} B$$

$$= 0.0266\,\frac{mv}{Bq} = 0.0266\ r$$

E